F79MB Chapter 8 GLM

Deviance

① Deviance for logistic regression.

Consider Binomial GLM with logistic link.

$D_i \sim Bin(n_i, \pi_i)$ with $\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 x_i$, for $i = 1, 2, \ldots, n$.

Then $L(\pi_i; d_i) = \binom{n_i}{d_i} \pi_i^{d_i} (1-\pi_i)^{n_i - d_i}$

$$\ell(\pi_i; d_i) = \log\binom{n_i}{d_i} + d_i \log(\pi_i) + (n_i - d_i)\log(1-\pi_i)$$

$$\ell(\underset{\sim}{\pi}; \underset{\sim}{d}) = \sum_{i=1}^{n} \left[\log\binom{n_i}{d_i} + d_i \log(\pi_i) + (n_i - d_i)\log(1-\pi_i)\right]$$

$$= c + \sum_{i=1}^{n} \left[d_i \log(\pi_i) + (n_i - d_i)\log(1-\pi_i)\right], \quad c = \text{constant}.$$

MLEs (for saturated model): $\frac{\partial \ell}{\partial \pi_i} = 0 \Rightarrow \boxed{\hat{\pi}_i = \frac{d_i}{n_i}}$.

$$\pi_i = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_i)}}$$

$$1 - \pi_i = 1 - \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_i)}} = \frac{e^{-(\beta_0 + \beta_1 x_i)}}{1 + e^{-(\beta_0 + \beta_1 x_i)}} = \frac{1}{1 + e^{\beta_0 + \beta_1 x_i}}$$

Model of Interest:

$$\ell(\underset{\sim}{\pi}, \underset{\sim}{d}) = c + \sum_{i=1}^{n} \left[d_i \log\left(\frac{\pi_i}{1-\pi_i}\right) + n_i \log(1-\pi_i)\right]$$

$$= c + \sum_{i=1}^{n} \left[d_i(\beta_0 + \beta_1 x_i) + n_i \log\left(\frac{1}{1 + e^{\beta_0 + \beta_1 x_i}}\right)\right]$$

$$= c + \sum_{i=1}^{n} d_i(\beta_0 + \beta_1 x_i) - \sum_{i=1}^{n} n_i \log(1 + e^{\beta_0 + \beta_1 x_i})$$

work out $\frac{\partial \ell}{\partial \beta_0}$ and $\frac{\partial \ell}{\partial \beta_1}$.

Solve numerically $\frac{\partial \ell}{\partial \beta_0} = \frac{\partial \ell}{\partial \beta_1} = 0$ for MLEs $\hat{\beta}_0, \hat{\beta}_1$.

(There is no analytical close form solution to find $\hat{\beta}_0, \hat{\beta}_1$)
We use numerical method instead.

$$\frac{\partial \ell}{\partial \beta_0} = \sum_{i=1}^{n} d_i - \sum_{i=1}^{n} n_i \cdot \frac{1}{1 + e^{\beta_0 + \beta_1 x_i}} \cdot e^{\beta_0 + \beta_1 x_i}(1) = 0$$

$$\frac{\partial \ell}{\partial \beta_1} = \sum_{i=1}^{n} d_i x_i - \sum_{i=1}^{n} n_i \cdot \frac{1}{1 + e^{\beta_0 + \beta_1 x_i}} \cdot e^{\beta_0 + \beta_1 x_i}(x_i) = 0$$

Denote $\hat{\eta}_i = \log\left(\frac{\hat{\pi}_i}{1-\hat{\pi}_i}\right) = \hat{\beta}_0 + \hat{\beta}_1 x_i$, fitted values for model of interest.

Residual Deviance

$$D = -2(\ell_\beta - \ell_{sat})$$

$$= -2\left(\left[\cancel{c} + \sum_{i=1}^{n} d_i \hat{\eta}_i - \sum_{i=1}^{n} n_i \log(1 + e^{\hat{\eta}_i})\right] - \left[\cancel{c} + \sum_{i=1}^{n} d_i \log\left(\frac{d_i}{n_i}\right) + \sum_{i=1}^{n} n_i \log\left(1 - \frac{d_i}{n_i}\right)\right]\right)$$

$$= -2\left[\sum_{i=1}^{n} d_i \hat{\eta}_i - \sum_{i=1}^{n} n_i \log(1 + e^{\hat{\eta}_i}) - \sum_{i=1}^{n} d_i \log\left(\frac{d_i}{n_i - d_i}\right) - \sum_{i=1}^{n} n_i \log\left(\frac{n_i - d_i}{n_i}\right)\right] \quad \#$$